Computer Architecture & Organisation.

* Computer organisation:-

→ Understanding the internal working of a computer.
→ Deciding which hardware will perform which task & how.
→ It includes:-
  → How data flows (between C.P.U / memory / I/O.)
  → How control signals are generated.
  → The structure of memory. (RAM, ROM, cache etc)
  → How the processor works and how instructions are executed.

* Computer Architecture:-

→ How the computer is designed.
→ It includes:-
(a) Instruction Set Architecture (ISA)
(b) Data type
(c) Register Design.
(d) Addressing mode.

Difference	organisation	Architecture
Focus	Implementation	Design
Related to user?	No	Yes
Example	Control signal, memory layout	Instruction set, Register
Feature Interested for	Hardware engineers, System designers.	Software developers, Computer designers

* Types of Computer Architecture:-

(1) Von-Neuman Architecture:-

→ The oldest and most widely used architecture in which data and instructions are accessed from the same memory.
→ Proposed by John von Neumann in 1945.

* Features:-

→ Single memory for program and data.
→ Instructions are executed sequentially.
→ Simple design
→ Program Counter tells the address of Instructions.

* Working:-

→ The CPU fetches instruction from memory.
→ Instructions are decoded by the C.U.
→ Instruction are executed by the ALU.
→ Results are stored in memory.

* Problem:-

→ Both data and Instructions share the same bus, so the CPU has to wait every time until data and instructions are fetched from memory.
→ Its cause performance to slow down.

ex: Intel 8085, 8086.

(2) Harvard Architecture:

→ Invented at Harvard University.
→ Data and instructions have seperate buses and memory.

* Working:

→ The CPU can read data from memory and fetch instructions from memory at the same time.
→ Since both tasks happen in parallel, the performance is fast.

* Features:

→ Seperate memory and buses.
→ Parallel processing possible

* Drawback:

→ Design - complex and costly.
→ Data and instruction have seperate memory space, which reduces flexibility.

Eg:- ARM cortex-M series, PIC controller, DSP

* Modified Harvard Architecture:

→ In modern CPUs, feature of both von Neuman and Harvard architecture are combined.
→ Memory is common (like von Neuman)
→ But there are seperate caches for data and instruction (like Harvard)
→ Provides higher speed and efficient design.

Eg:- Intel i7, AMD Ryzen, ARM Cortex A Series.

(5) ISA (Instruction Set Architecture).

→ The interface b/w hardware (c.p.u) and software (program).
→ Defines which instructions the processor can understand and executes.

* Types of ISA:-

(A) CISC:-

→ Complex instruction Set Computing.
→ It has many complex instructions
→ One instruction can perform multiple low-level operations.
→ Example: Intel X86.
→ Its code size is small because fewer instructions need to be written.
→ The main disadvantage is that its execution speed is slow due to complex decoding.

(B) RISC:-

→ Reduced instruction Set Computing.
→ Simple and fixed length instructions.
→ Each instruction executes in one clock cycle.
→ eg:- ARM.
→ Faster execution.
→ Easy to design pipeline.
→ main advantage: code size becomes very large.

* Types of instructions in DSA :-

(i) Data Transfer : (MOV, LOAD, STORE)
(ii) Arithmetic : (ADD, SUB, MUL)
(iii) Logical :- (AND, OR, AND)
(iv) Control flow :- (JMP, CALL, RET)
(v) Input/output :- (IN, OUT)
(vi) Comparison :- (CMP)

* Pipeline :-

→ Technique that improves the performance of CPU.
→ Instruction processed step by step.
→ Normal CPU (without pipeline) complete one instruction and then start the next.
→ But in pipeline, while one instruction is being decoded, the next instruction can be fetched.
→ Parallel execution is allowed.
→ with pipelining, CPU speed increased and throughput also increased.

(4) Microarchitecture :-

→ This architecture shows how the internal structure of the CPU is built and how its components work internally.
→ If system architecture is a map, then microarchitecture is the detailed information of that map.

Main Components of microarchitecture :-

(i) Register :-
→ fastest, smallest and costliest memory of the Computer.
→ Example :- Program Counter, Accumulator ...
→ Inbuilt memory of the CPU.

(ii) ALU

(iii) Control Unit :-
→ Controls & manages all the devices & resource, Which is
Why it is known as the Central nervous System.
→ Decodes instructions and generates timing & control signals.

(iv) Cache :-
→ Buffer memory between CPU & RAM.
→ Manages Speed between CPU & RAM
→ 4 types :- L1, L2, L3, L4 (L1 fastest)

(v) Pipeline :-
→ Technique to process multiple instruction at the same time.
→ Fetch → Decode → Execute → write Back.

(5) System design Architecture :-
→ Shows the arrangement of the System's major components and the data flow b/w them.
→ High level design that shows how the System will work and how its major units connected.

* Error detection & correction codes:-

Types:

(1) Single Bit error :-

→ only one bit gets changed.

→ 101011 → 101001

(2) Burst Error :-

→ multiple bits get changed.

* Error detection codes:-

(i) Parity bit method :- (only to detect single bit error)

→ An extra bit (Parity bit) is transmitted along with the data.

→ Total no of 1's either even or odd.

Types:

(a) Even Parity:

→ no of 1's must be even.

Ex: 101001 (Total 3 ones)

Add parity bit :- 1010011 (even)

(b) odd Parity:

→ no. of 1's must be odd.

→ Data bit : 101001

↓

1010010 (1's already odd)

(ii) CheckSum method :-

→ mostly used in network data transmission.
→ Data is divided into equal-sized block / segments.
→ The sum of all blocks is generated using a CheckSum.
→ This CheckSum is also checked at the receiver end.

Ex: Data blocks :- 1 = 1001
2 = 1100
3 = 0101

Sum = 1001 + 1100 + 0101 = 1 1010 (even)
↓ carry ignore

⇒ 1010
sender.

* Error Correction Code :-

→ To detect errors and automatically correct them, without data retransmission.
→ used in more reliable system like memory, satellite comm. etc

(i) Hamming Code :-

→ Proposed by Richard Hamming.
→ Performs both single-bit error detection and correction
→ In this code, some redundent bit (r) are added among the data bits (d), so that each bit position can be represented by a unique binary number.
→ To find the redundent bits :- $2^r \geq (r+d+1)$

eg: 4 data bits :- $2^r \geq (r+4+1) \Rightarrow \boxed{r=3}$

5

* Bus :-

→ A communication path / Set of wires / Set of lines, through which, in a computer system, data, address, and control signals are exchanged b/w multiple parts.

Types:

(i) Address Bus :-

→ Unidirectional
→ The bus carries the address of the memory location where data is to be read/written.
→ When the CPU reads data from memory, it sends the address of that memory location through the address bus.
→ Example :- If the CPU uses a 16-bit address bus, it can access $2^{16} = 65536$ (64KB) mem location

(ii) Data Bus :-

→ Bidirectional
→ This bus carries the actual data being transferred b/w the CPU, memory and I/O.
→ CPU → memory (Data write)
→ CPU ← memory (Data read)
Ex: If the CPU has a data bus width of 16 bits, then it can carry 2 bytes of data per cycle.

(c) Control Bus:-

→ Carries Control and timing Signals.
→ This bus indicates which type of operation is being performed (Read / write / Interrupt)
→ Direction = Both (Based on Control Signal).

* Microprocessor:-

→ Invention = Year 1971
→ Intel 4004 by Intel Incorporation.
→ 1st Smallest & Complete CPU (4 bits)
→ After 1971, Intel 8008, Intel 80386 etc. processor came, and Intel i Series (Itenium) & AMD Ryzen Series Complex & multitasking CPUs are available.
→ Use = Personal Computers, Laptops, Servers, Gaming Console (PlayStations & Xbox)
→ Intel X86 Came in 1978 (based on Intel 8086 Architecture)

Types:-

(a) RISC
(b) CISC
(c) VLIW (Very long Instruction word) (Parallel processing)
(d) EPIC (Explicitly parallel Instruction Computing) (Advanced VLIW)

* Micro Controller :-

→ Along with the microprocessor, memory (RAM & ROM) and Peripherals (Input-output & timers) are also used, so it is mainly used in Embedded Systems.

→ first microcontroller = Intel 8051 (1980)

→ After this, PIC, ARM, AVR are popular microcontrollers

→ Use = Embedded System (microwave oven, washing machine, Robotics, Automice, Automobiles, medical devices.

Types :-

(a) PIC :-

→ Peripheral Interface Controller.

→ Developed by :- Microchip Technology.

→ Size : 8, 16, 32-bits variants.

→ Use :- Consumer Electricity, Embedded System & Automotive

(b) AVR :-

→ Advanced virtual RISC

→ Developed by :- Atmel (now microchip)

→ Based on RISC

→ used in small embedded devices.

(c) ARM :-

→ Advanced RISC machine.

→ RISC-based powerful microcontrollers family.

→ well known for low power consumption yet high performance

→ widely used in smart phones & Embedded Systems

Comparision Point	Parity bit method	Check Sum method	Hamming Code
Definition	Simple error detection by adding single bit parity	Data block added, 1s complement, Checksum generated	Uses data + redundant parity bits for single bit error correction
main goal	Error detection	Error detection	Error detection + error correction
Category	Error detection code	Error detection code	Error correction code
Added Redundant Bits	only 1 bit	16 or 32 bit Checksum	Multiple parity bits (3, 4, 5 depend on data size)
placement of Extra bits	End of data frame	After data, as Checksum block	Parity bit at position 1, 2, 4, 8
Technique used	Count 1's = Even or odd	Binary addition + wrap around carry	Parity bits calculated using multiple overlapped bit groups.
Detects single bit errors	Yes	Yes	Yes
Correct single bit	NO	NO	Yes
Detect double bit errors	NO	No (Sometimes)	Yes (detect only not correct by Hamming & BCH code)
Reliability	Low	medium	very High
Use	Simple serial communication	TCP, UDP, IP, Ethernet	ECC RAM, Satellite
Pros	Simple, Cheap, fast	large error detection	Detect + correction = very accurate
Cons	only detection	No correction	Complex & more overhead
use case.	Low risk	Networking protocols	High reliability systems.